EXPRIMENT-2

2: Create labelled matrices of sizes 5×4, 3×3, and 2×2 in R, filling them by row and by column.

PROGRAM:

```
# Create a 5x4 matrix filled by row
m1 <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE)
print("5x4 Matrix")
print(m1)

# Create a 3x3 matrix filled by column
m2 <- matrix(1:9, nrow=3, ncol=3, byrow=FALSE)
print("3x3 Matrix")
print(m2)

# Create a 2x2 matrix with row and column labels
m3 <- matrix(1:4, nrow=2, ncol=2, byrow=TRUE,
             dimnames=list(c("Row1", "Row2"),
                           c("Col1", "Col2")))
print("2x2 Labelled Matrix")
print(m3)

```

OUTPUT:

```
Console   Terminal   Background Jobs
R 4.6.1 · ~/
     [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
>
> # Create a 2x2 matrix with row and column labels
> m3 <- matrix(1:4, nrow=2, ncol=2, byrow=TRUE,
+              dimnames=list(c("Row1", "Row2"),
+                            c("Col1", "Col2")))
> print("2x2 Labelled Matrix")
[1] "2x2 Labelled Matrix"
> print(m3)
     Col1 Col2
Row1    1    2
Row2    3    4
>
```